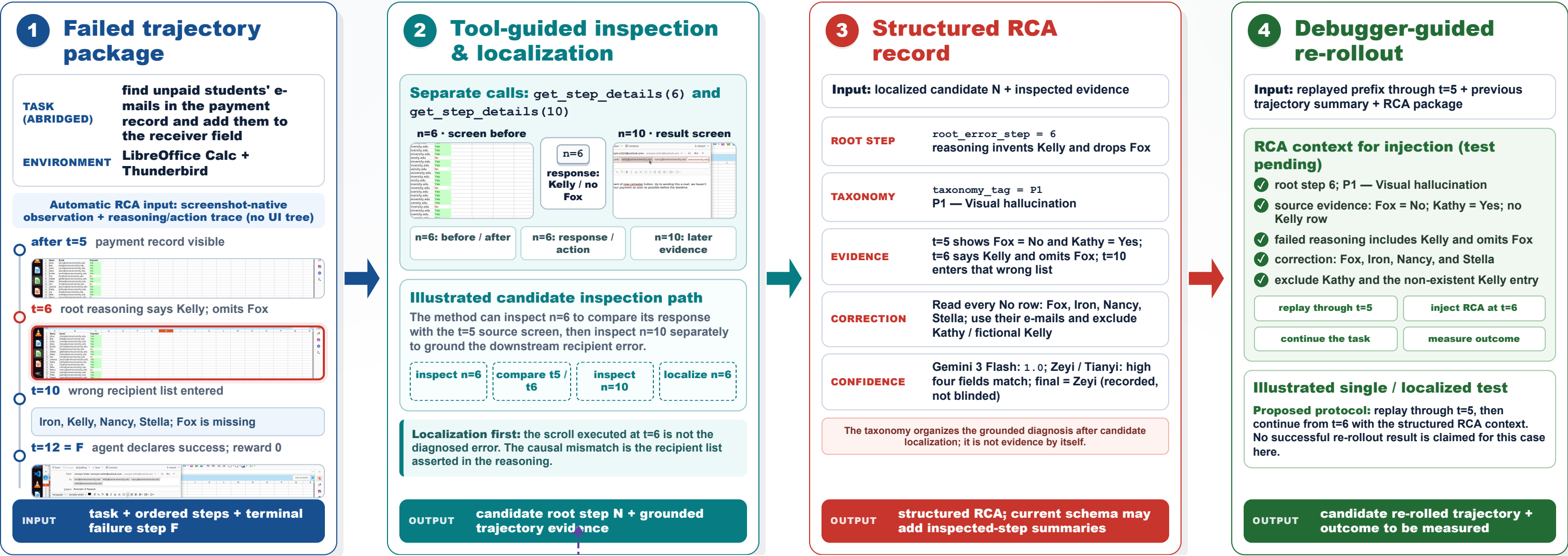


CUADebugger Pipeline

Inspect screenshot-native failures, localize the earliest causal mismatch, emit structured RCA, and support a controlled re-rollout test.



4

Debugger-guided re-rollout

Input: replayed prefix through t=5 + previous trajectory summary + RCA package

RCA context for injection (test pending)

✓ root step 6; P1 — Visual hallucination

✓ source evidence: Fox = No; Kathy = Yes; no Kelly row

✓ failed reasoning includes Kelly and omits Fox

✓ correction: Fox, Iron, Nancy, and Stella

✓ exclude Kathy and the non-existent Kelly entry

replay through t=5

inject RCA at t=6

continue the task

measure outcome

Illustrated single / localized test

Proposed protocol: replay through t=5, then continue from t=6 with the structured RCA context. No successful re-rollout result is claimed for this case here.

OUTPUT

candidate re-rolled trajectory + outcome to be measured

Optional Episodic Memory — candidate RCA context; memory-augmented RCA guides re-rollout

Retrieval is semantic and can reuse failure patterns across the current multi-application pool; held-out-application transfer is not established.

1 • Distill prior failures

Store trigger, failed action, correction, tag, distinguishing feature, evidence, and an episodic reference.

2 • Build a query

Use task, app, observed state, suspected action, or tentative tag as retrieval cues.

3 • Retrieve candidates

Return top-k lessons; follow an episodic reference when source evidence is needed.

4 • Verify before use

Check trigger and distinguishing feature against screenshots and trace; discard mismatches.

INTERPRETATION BOUNDARY

Staged component evidence, not a factorial ablation. Continual re-rollout is the clean fixed-control comparison; single re-rollout is a package comparison because restart source and prompt format differ.